Ferry Capacity Utilization & Operational Efficiency Analytics System

Research Paper: Exploratory Data Analysis, Insights & Recommendations

Jack Layton Ferry Terminal — Toronto Island Ferries
Prepared for: Toronto Government Parks, Forestry & Recreation
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Abstract

This research paper presents an operational analytics study of ferry capacity utilization at the Jack Layton Ferry Terminal, based on 261,538 fifteen-minute ticket sales and redemption records spanning May 2015 to 2025. Rather than forecasting future demand, this study quantifies how efficiently existing ferry capacity is utilized, identifying recurring congestion and idle-capacity windows to support evidence-based operational planning. Key findings include a dominant seasonal congestion pattern (94.3% of all sustained high-congestion events occur in summer months despite summer representing only 34.6% of operating days), a validated single-day extreme event (a 13-hour continuous congestion period on Canada Day 2018), and an average capacity utilization of 9.8% across all measured intervals — a figure that masks substantial seasonal and time-of-day variation explored in detail below.

1. Introduction & Problem Statement

Ferries operating from the Jack Layton Ferry Terminal provide year-round transportation to Centre Island, Hanlan's Point, and Ward's Island. Ferry operations are constrained by fixed vessel capacity, staffing limits, and highly seasonal demand fluctuations. Despite the availability of granular ticket sales and redemption data, ferry operations have historically lacked a structured analytical framework to measure capacity utilization patterns, identify over- and under-utilization windows, quantify operational inefficiencies during off-peak periods, and support evidence-based resource allocation.

This study addresses that gap through a pure operational analytics lens — deliberately distinct from demand forecasting — with the following objectives:

Primary Objectives

- Quantify ferry capacity utilization using ticket activity data
- Identify inefficiencies in ferry deployment across time intervals
- Detect congestion-prone and idle-capacity periods

Secondary Objectives

- Support operational cost optimization
- Improve passenger comfort and safety
- Enable strategic planning for seasonal operations

2. Data Description

The dataset consists of 261,538 records at 15-minute interval resolution, spanning May 2015 through 2025, with the following fields:

Column	Description
_id	Unique row identifier
Timestamp	15-minute interval end time
Sales Count	Tickets sold in the interval
Redemption Count	Tickets redeemed (boarded) in the interval

Sales and Redemption counts are not necessarily synchronized within the same interval: tickets may be purchased in advance (online or in bulk for groups) and redeemed at a later date or time, a pattern that explains a number of high-Sales, low-Redemption intervals observed during data validation (Section 3.3).

3. Methodology

3.1 Data Cleaning & Validation

The raw dataset was processed through the following validation steps:

- Timestamp parsing and conversion to datetime format, sorted chronologically
- Duplicate timestamp check — none found (0 duplicates)
- Negative/invalid value check — minimum Sales and Redemption counts both confirmed at 0; no negative values present
- Gap analysis within active service days — conducted to distinguish genuine data gaps from normal non-operating hours (overnight/off-season), avoiding the common pitfall of treating scheduled non-operating time as missing data
- Extreme value flagging using the 99th percentile threshold per column (Sales: 458; Redemption: 508), flagging 2,598 sales spikes and 2,609 redemption spikes for downstream review rather than removal

3.2 Feature Engineering

The following derived features were constructed to enable capacity and efficiency analysis:

Feature	Definition
Total Activity Load	Sales Count + Redemption Count
Redemption Pressure Ratio	Redemption Count / (Sales Count + 1)
Capacity Utilization Ratio	Total Activity Load / Assumed Maximum Interval Capacity
Operational Load Index (OLI)	Rolling 1-day (96-interval) min-max normalized activity score, 0–1
Idle Capacity Indicator	Flag for ≥4 consecutive intervals (≥1 hour) below the 10th percentile of activity
Congestion Indicator	Flag for ≥2 consecutive intervals (≥30 min) above the 90th percentile of activity

3.3 Capacity Assumption — Methodological Note

The dataset does not include vessel-level capacity figures. An assumed maximum interval capacity was therefore required to compute the Capacity Utilization Ratio. This value was calibrated empirically against the observed distribution of Total Activity Load rather than selected arbitrarily:

Percentile / Statistic	Total Activity Load Value
95th percentile	494
99th percentile	962
Maximum observed value	14,445

The assumed maximum interval capacity was set at 1,000 ticket events (sales + redemptions combined), placed just above the 99th percentile (962). This reflects a realistic upper bound of near-maximum vessel and staffing throughput while avoiding distortion from a small number of extreme outlier intervals (see Section 3.4). This assumption should be revisited and refined with actual vessel capacity specifications if they become available; it is disclosed transparently here as a methodological limitation.

3.4 Anomaly Investigation

Forty-seven intervals (approximately 0.02% of the dataset) recorded Total Activity Load values exceeding 2,000 — more than double the 99th percentile and, in the most extreme case, reaching 14,445 (15× the 99th percentile). Two distinct sub-patterns were identified within this group:

- Intervals with very high Sales counts and near-zero Redemption counts in the same interval (e.g., 2015-05-19: 3,017 sales, 0 redemptions) — consistent with bulk or advance ticket purchases (e.g., group/event bookings) where redemption occurs across many later, separate intervals.
- Intervals where Sales and Redemption counts are nearly identical within the same window (e.g., 2023-08-15 20:15: 7,229 sales, 7,216 redemptions) — consistent with same-day, walk-up visitor activity where purchase and boarding occur in immediate succession.

Both patterns are consistent with plausible real-world ticketing behavior rather than data entry error, and were therefore retained in the analysis rather than removed. They are flagged for stakeholder awareness as they represent the extreme tail of demand and disproportionately influence maximum-value statistics.

4. Key Performance Indicator Results

The following KPIs were computed across the full 10-year dataset (261,538 intervals):

KPI	Result
Capacity Utilization Ratio	0.098 (9.8%)
Congestion Pressure Index	8.25%
Idle Capacity Percentage	0.46%
Peak Strain Duration (longest sustained congestion streak)	795 minutes (≈13.25 hours)
Operational Variability Score (coefficient of variation)	2.033

4.1 Interpretation

- Capacity Utilization Ratio (9.8%): This is a full-dataset average spanning all hours of all days, including off-season overnight periods with minimal activity. It should not be read as “the ferry is underused” in isolation — Section 5 demonstrates that utilization is highly concentrated in specific seasonal and time-of-day windows rather than evenly distributed.
- Congestion Pressure Index (8.25%): Approximately one in every twelve 15-minute intervals qualifies as a sustained congestion period, indicating a meaningful and recurring operational strain rather than rare, isolated events.
- Idle Capacity Percentage (0.46%): Sustained idle periods (≥1 hour of very low activity) are rare across intervals where the ferry is actively running, suggesting that scheduled service hours are, for the most part, not significantly over-provisioned relative to demand.

- Peak Strain Duration (795 minutes): The single longest uninterrupted high-congestion event in 10 years lasted approximately 13 hours — identified in Section 5.1 as occurring on July 1, 2018 (Canada Day).
- Operational Variability Score (2.033): A coefficient of variation above 2 indicates highly volatile, “spiky” demand relative to the mean — consistent with a leisure/tourism-driven ferry system rather than a steady commuter-pattern system.

5. EDA Findings & Insights

5.1 Case Study: The Canada Day 2018 Strain Event

The longest sustained congestion streak identified in the dataset (53 consecutive 15-minute intervals, 795 minutes) occurred on July 1, 2018 — Canada Day. Detailed interval-level inspection shows a clear demand curve:

- 08:00–11:00: Ramp-up phase, activity rising from ~300 to ~1,150 total activity load per interval
- 11:00–18:00: Sustained peak plateau, consistently between 1,000–1,350 per interval for approximately 7 continuous hours
- 18:00–21:00: Gradual decline back to baseline (~300) as evening departures tapered off

This single-day event represents a textbook holiday demand curve and provides direct, dateable evidence for holiday-specific operational planning (Section 6).

5.2 Seasonal Concentration of Congestion (Validated Finding)

Analysis of all 596 high-congestion events (defined as streaks ≥20 consecutive intervals) across the 10-year dataset reveals a strong seasonal concentration:

Metric	Result
High-congestion days in summer months (Jun–Sep)	94.3%
High-congestion days on weekends (any month)	46.3%
High-congestion days that are both summer AND weekend	41.4%
High-congestion days that are summer weekdays (non-weekend)	52.9%

To rule out the possibility that this concentration simply reflects more operating days existing in summer, operating-day counts were compared directly:

Period	Unique Operating Days
Summer (June–September)	1,342 days (34.6% of all operating days)
Off-season (October–May)	2,540 days (65.4% of all operating days)

Summer represents only 34.6% of all operating days in the dataset, yet accounts for 94.3% of all sustained high-congestion events — a disproportion that confirms a genuine seasonal demand effect rather than a data-availability artifact. Notably, summer weekdays (52.9%) slightly outnumber summer weekends (41.4%) among high-congestion days, indicating that congestion risk extends across the full week during peak season and is not concentrated on weekends alone. This is consistent with a tourism-driven ridership base, where visitor patterns are not constrained to a Saturday/Sunday cadence the way local commuter traffic would be.

5.3 Weekday, Time-of-Day, and Seasonal Activity Patterns

Supplementary breakdowns (full charts available in the accompanying notebook and dashboard) show:

- Weekend intervals show substantially higher average Total Activity Load than weekday intervals, consistent with general leisure-travel expectations, even though Section 5.2 shows this weekend effect is secondary to the seasonal effect for sustained congestion specifically.
- Summer (Peak) season shows roughly 5–6× higher average activity than the off-season, reinforcing the seasonal concentration finding above.
- Afternoon hours show the highest average activity load among the four time-of-day bands (Morning, Afternoon, Evening, Night/Off-hours), followed by Morning, then Evening, with Night/Off-hours activity near zero as expected for non-operating hours.

6. Recommendations

6.1 Season-Long Capacity Planning (Primary Recommendation)

Given that 94.3% of sustained high-congestion events occur within the four summer months, and that summer weekdays contribute as much congestion risk as summer weekends, it is recommended that additional vessel deployment, staffing, and crowd-management resources be allocated on a season-long basis across the entire June–September window — rather than scheduled narrowly around weekends or isolated special-event dates.

6.2 Holiday-Specific Reinforcement

Independent of the broader seasonal pattern, statutory holidays such as Canada Day (July 1) show some of the most extreme single-day congestion events in the dataset (Section 5.1), appearing as elevated-demand days regardless of which weekday they fall on across multiple years. Pre-positioning additional capacity specifically on these calendar dates, on top of standard summer staffing, is recommended.

6.3 Off-Season Efficiency Review

With idle capacity periods comprising only 0.46% of all intervals, there is limited evidence of broad over-provisioning during scheduled service hours. Any cost-optimization review should focus on the structure of off-season scheduled hours themselves (e.g., whether all currently scheduled off-season service windows are necessary) rather than on trimming service during already-lean operating periods.

6.4 Capacity Assumption Refinement

The Capacity Utilization Ratio KPI is dependent on an assumed maximum interval capacity (1,000 ticket events) that was estimated statistically rather than sourced from vessel specifications. It is recommended that Toronto Parks, Forestry & Recreation provide actual vessel-level capacity figures (by vessel type: Ongiara, Sam McBride, Thomas Rennie, William Inglis, Trillium) so this KPI can be recalculated with operational precision.

6.5 Continued Monitoring of Extreme-Demand Intervals

The 47 intervals identified with anomalously high activity (Section 3.4) should be reviewed by operations staff with knowledge of specific dates (e.g., known large group bookings, festivals, fireworks events) to confirm the bulk-purchase and same-day-surge interpretations proposed in this report, and to inform future advance-capacity planning for similar events.

7. Limitations

- Vessel-level capacity data was not available in the source dataset; the Capacity Utilization Ratio relies on a statistically-derived assumption (Section 3.3) rather than ground-truth vessel specifications.
- This study is restricted to operational efficiency analysis and does not incorporate weather data, special event calendars, or external tourism statistics, which likely contribute to the seasonal and holiday patterns identified but were outside the scope of available data.
- The 2020–2021 period falls within the dataset window and reflects known COVID-19 service disruptions; trend analyses spanning this period should account for this external, non-operational disruption rather than attributing changes to normal demand variation.
- Idle and congestion thresholds are defined using percentile-based statistical cutoffs (10th/90th percentile) rather than fixed operational definitions provided by the terminal operator; results should be interpreted as relative, data-driven indicators rather than absolute capacity breach thresholds.

8. Conclusion

This study demonstrates that ferry terminal congestion at Jack Layton Ferry Terminal is driven predominantly by seasonal demand rather than by day-of-week scheduling patterns, with summer months accounting for a disproportionate 94.3% of all sustained high-congestion events despite representing only 34.6% of operating days. A single extreme case study — the 13-hour continuous congestion event on Canada Day 2018 — further illustrates the scale of demand concentration possible on individual high-demand dates. These findings, derived entirely from ticket sales and redemption data without reliance on demand forecasting models, provide Toronto Government Parks, Forestry & Recreation with a concrete, evidence-based foundation for shifting from static, calendar-based scheduling toward season-long, data-informed operational planning.